



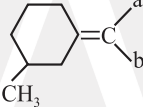
## CHAPTER

## 9

# Organic Chemistry- Some basic Principles and Techniques

| Sr. No.              | Concept                                                                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Other information |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
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| 1.                   | IUPAC nomenclature:                                                                                                                                | <table><tr><th>Functional group</th><th>Structure</th><th>Prefix</th><th>Suffix</th></tr><tr><td>Carboxylic acid</td><td><math>\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OH} \end{array}</math></td><td>Carboxy</td><td>– oic acid</td></tr><tr><td>Sulphonic acid</td><td><math>-\text{SO}_3\text{H}</math></td><td>Sulpho</td><td>Sulphonic acid</td></tr><tr><td>Anhydride</td><td><math>\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \quad \text{O} \\ \parallel \\ -\text{C} \quad \text{O} \\ \parallel \\ \text{O} \end{array}</math></td><td>×</td><td>Oic-anhydride</td></tr><tr><td>Ester</td><td><math>\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OR} \end{array}</math></td><td>Alkoxy carbonyl or Carbalkoxy</td><td>Alkyl .... oate</td></tr><tr><td>Acid chloride</td><td><math>\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{Cl} \end{array}</math></td><td>Chloroformyl or Chlorocarbonyl</td><td>– oyl chloride</td></tr><tr><td>Acid amide</td><td><math>\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{NH}_2 \end{array}</math></td><td>Carbamoyl/ Amido</td><td>–amide</td></tr><tr><td>Carbonitrile/Cyanide</td><td><math>-\text{C}\equiv\text{N}</math></td><td>Cyano</td><td>nitrile</td></tr><tr><td>Aldehyde</td><td><math>\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{H} \end{array}</math></td><td>Formyl or Oxo</td><td>– al</td></tr><tr><td>Ketone</td><td><math>\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}- \end{array}</math></td><td>Keto or oxo</td><td>– one</td></tr><tr><td>Alcohol</td><td><math>-\text{OH}</math></td><td>Hydroxy</td><td>– ol</td></tr><tr><td>Thio alcohol</td><td><math>-\text{SH}</math></td><td>Mercapto</td><td>thiol</td></tr><tr><td>Amine</td><td><math>-\text{NH}_2</math></td><td>Amino</td><td>amine</td></tr><tr><td>Ether</td><td><math>-\text{O}-\text{R}</math></td><td>Alkoxy</td><td>–</td></tr><tr><td>Oxirane</td><td></td><td>Epoxy</td><td>–</td></tr><tr><td>Nitro derivative</td><td><math>-\text{NO}_2</math></td><td>Nitro</td><td>–</td></tr><tr><td>Nitroso derivative</td><td><math>-\text{NO}</math></td><td>Nitroso</td><td>–</td></tr><tr><td>Halide</td><td><math>-\text{X}</math></td><td>Halo</td><td>–</td></tr><tr><td>Double bond</td><td><math>\text{C} = \text{C}</math></td><td>–</td><td>ene</td></tr><tr><td>Triple bond</td><td><math>\text{C} \equiv \text{C}</math></td><td>–</td><td>yne</td></tr></table> | Functional group  | Structure | Prefix | Suffix | Carboxylic acid | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OH} \end{array}$ | Carboxy | – oic acid | Sulphonic acid | $-\text{SO}_3\text{H}$ | Sulpho | Sulphonic acid | Anhydride | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \quad \text{O} \\ \parallel \\ -\text{C} \quad \text{O} \\ \parallel \\ \text{O} \end{array}$ | × | Oic-anhydride | Ester | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OR} \end{array}$ | Alkoxy carbonyl or Carbalkoxy | Alkyl .... oate | Acid chloride | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{Cl} \end{array}$ | Chloroformyl or Chlorocarbonyl | – oyl chloride | Acid amide | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{NH}_2 \end{array}$ | Carbamoyl/ Amido | –amide | Carbonitrile/Cyanide | $-\text{C}\equiv\text{N}$ | Cyano | nitrile | Aldehyde | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{H} \end{array}$ | Formyl or Oxo | – al | Ketone | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}- \end{array}$ | Keto or oxo | – one | Alcohol | $-\text{OH}$ | Hydroxy | – ol | Thio alcohol | $-\text{SH}$ | Mercapto | thiol | Amine | $-\text{NH}_2$ | Amino | amine | Ether | $-\text{O}-\text{R}$ | Alkoxy | – | Oxirane |  | Epoxy | – | Nitro derivative | $-\text{NO}_2$ | Nitro | – | Nitroso derivative | $-\text{NO}$ | Nitroso | – | Halide | $-\text{X}$ | Halo | – | Double bond | $\text{C} = \text{C}$ | – | ene | Triple bond | $\text{C} \equiv \text{C}$ | – | yne |  |
| Functional group     | Structure                                                                                                                                          | Prefix                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Suffix            |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Carboxylic acid      | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OH} \end{array}$                                                                        | Carboxy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | – oic acid        |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Sulphonic acid       | $-\text{SO}_3\text{H}$                                                                                                                             | Sulpho                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sulphonic acid    |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Anhydride            | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C} \quad \text{O} \\ \parallel \\ -\text{C} \quad \text{O} \\ \parallel \\ \text{O} \end{array}$ | ×                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Oic-anhydride     |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Ester                | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{OR} \end{array}$                                                                        | Alkoxy carbonyl or Carbalkoxy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Alkyl .... oate   |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Acid chloride        | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{Cl} \end{array}$                                                                        | Chloroformyl or Chlorocarbonyl                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | – oyl chloride    |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Acid amide           | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{NH}_2 \end{array}$                                                                      | Carbamoyl/ Amido                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | –amide            |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Carbonitrile/Cyanide | $-\text{C}\equiv\text{N}$                                                                                                                          | Cyano                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | nitrile           |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Aldehyde             | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}-\text{H} \end{array}$                                                                         | Formyl or Oxo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | – al              |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Ketone               | $\begin{array}{c} \text{O} \\ \parallel \\ -\text{C}- \end{array}$                                                                                 | Keto or oxo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | – one             |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Alcohol              | $-\text{OH}$                                                                                                                                       | Hydroxy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | – ol              |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Thio alcohol         | $-\text{SH}$                                                                                                                                       | Mercapto                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | thiol             |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Amine                | $-\text{NH}_2$                                                                                                                                     | Amino                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | amine             |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Ether                | $-\text{O}-\text{R}$                                                                                                                               | Alkoxy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | –                 |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Oxirane              |                                                                                                                                                    | Epoxy                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | –                 |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Nitro derivative     | $-\text{NO}_2$                                                                                                                                     | Nitro                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | –                 |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Nitroso derivative   | $-\text{NO}$                                                                                                                                       | Nitroso                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | –                 |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Halide               | $-\text{X}$                                                                                                                                        | Halo                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | –                 |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Double bond          | $\text{C} = \text{C}$                                                                                                                              | –                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | ene               |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |
| Triple bond          | $\text{C} \equiv \text{C}$                                                                                                                         | –                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | yne               |           |        |        |                 |                                                                             |         |            |                |                        |        |                |           |                                                                                                                                                    |   |               |       |                                                                             |                               |                 |               |                                                                             |                                |                |            |                                                                               |                  |        |                      |                           |       |         |          |                                                                            |               |      |        |                                                                    |             |       |         |              |         |      |              |              |          |       |       |                |       |       |       |                      |        |   |         |  |       |   |                  |                |       |   |                    |              |         |   |        |             |      |   |             |                       |   |     |             |                            |   |     |  |

| Sr. No. | Concept                                                                                                              | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Other information |
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| 2.      | Nomenclature of organic compounds                                                                                    | <p>IUPAC name = 2° prefix + 1° prefix + word root + 1° suffix + 2° suffix</p> <div style="display: flex; justify-content: space-between;"> <div style="width: 30%;"> <p>Some functional groups are always treated as substituents whether organic compound is mono or polyfunctional e.g., Fluoro (-F), Chloro (-Cl), Bromo (-Br), Iodo (-I), Nitro (-NO<sub>2</sub>), Nitroso (-NO), Diazo (-N=N), Alkoxy (-OR), Alkyl (-R), Phenyl(-C<sub>6</sub>H<sub>5</sub>), etc.</p> <p>When functional groups act as substituents in the presence of other principal functional group e.g., Hydroxyl (-OH), Formyl (-CHO), Oxo (&gt;CO), Cyano (-CN), Carboxy (-COOH), Alkoxycarbonyl (-COOR), Halocarbonyl (COX), Amino (-NH<sub>2</sub>), Carbamoyl (-CONH<sub>2</sub>), Sulpho (-SO<sub>3</sub>H), etc.</p> </div> <div style="width: 30%;"> <p><b>Principal chain</b></p> <p>If no. of C atoms is C<sub>1</sub>-meth, C<sub>2</sub>-eth, C<sub>3</sub>-prop(a), C<sub>4</sub>-but(a) C<sub>5</sub>-pent(a), C<sub>6</sub>-hex(a) C<sub>7</sub>-hept(a), C<sub>8</sub>-oct(a) C<sub>9</sub>-non(a) C<sub>10</sub>-dec(a) C<sub>11</sub>-undec(a), C<sub>12</sub>-dodec(a). Extra 'd' is added to word root if 1 suffix begins with a consonant.</p> <p>Cyclo is used for carbocyclic compounds. If cyclo is not put, it indicates acyclic or open chain compound. In cyclic compound, a more branched carbon gets lower number.</p> </div> <div style="width: 30%;"> <p>Indicates functional group.<br/>-OH(-ol), -CHO(-al), &gt;CO(-one), -COOH(-oic acid), -CONH<sub>2</sub>(-amide), -COCl(-oyl chloride), -COOR(alkyl.....oate), -CN(nitrile), -SH(thiol), -NH<sub>2</sub>(amine), -SO<sub>3</sub>H(sulphonic acid).<br/>If 2° suffix begins with a vowel, the terminal e is dropped otherwise it is retained</p> <p>Single bonds (-) only-ane<br/>1(=)bond-ene<br/>2(=)bonds-diene<br/>3(=)bonds-triene<br/>1(≡)bond-yne<br/>2(≡)bonds-diynes</p> </div> </div> <p>Preference order (decreasing priority) for selection of a principal functional group in polyfunctional compounds is<br/>-COOH, -SO<sub>3</sub>H, -COOR, -COCl, -CONH<sub>2</sub>, -CN, -HC=O<br/>&gt;C=O, -OH, -NH<sub>2</sub>, &gt;C=C&lt;, -C≡C-</p> |                   |
| 3.      | Nomenclature of substituted benzene compounds                                                                        | <p>(1, 2-) ortho      (1, 3-) meta      (1, 4-) para</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |
| 4.      | Isomerism<br>(a) Constitutional Isomerism/Structural isomerism<br>(i) Chain isomerism<br><br>(ii) Position isomerism | <p>Having same molecular formula but different structures.</p> <p>Having different carbon skeletons.</p> <p>e.g. C<sub>5</sub>H<sub>12</sub> : CH<sub>3</sub>-CH<sub>2</sub>-CH<sub>2</sub>-CH<sub>2</sub>-CH<sub>3</sub><br/>n-pentane</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>iso-pentane</p> </div> <div style="text-align: center;"> <p>neo-pentane</p> </div> </div> <p>Different position of functional group on the carbon chain.</p> <div style="display: flex; justify-content: space-around;"> <div style="text-align: center;"> <p>o-xylene</p> </div> <div style="text-align: center;"> <p>m-xylene</p> </div> <div style="text-align: center;"> <p>p-xylene</p> </div> </div>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |

| Sr. No. | Concept                    | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Other information |
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|         | (iii) Functional isomerism | <p>Having different functional group.<br/>e.g. <math>C_3H_9N</math> : <math>CH_3 - CH_2 - CH_2 - NH_2</math>,<br/>1-propanamine</p> <p><math>CH_3 - CH_2 - NH - CH_3</math>,<br/>N-methylethanamine</p> <p><math>\begin{array}{c} CH_3 \\   \\ CH_3 - N - CH_3 \end{array}</math>,<br/>N, N-dimethylmethanamine</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                   |
|         | (iv) Metamerism            | <p>Having different alkyl groups on either side of the functional groups.<br/>e.g. <math>C_4H_{10}O</math> : <math>CH_3 - O - CH_2 - CH_2 - CH_3</math><br/>n-propyl methylether</p> <p><math>CH_3 - CH_2 - O - CH_2 - CH_3</math><br/>diethyl ether</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |
|         | (v) Tautomerism            | <p>Arises due to migration of a H-atom from one polyvalent atom to the other within the same molecule with necessary rearrangement of linkage.<br/>e.g. Acetoacetic ester.</p> <p><math>\begin{array}{c} O \\    \\ CH_3 - C - CH_2COOC_2H_5 \end{array} \rightleftharpoons \begin{array}{c} OH \\   \\ CH_3 - C = CHCOOC_2H_5 \end{array}</math><br/>keto form                      enol form</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                   |
| (b)     | Stereoisomerism            | <p>Having same molecular formula &amp; structures but different configuration, i.e. arrangement of their atoms in space.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                   |
|         | (i) Geometrical isomerism  | <p><math>\begin{array}{c} a &amp; &amp; a \\ &amp; \diagdown &amp; / \\ &amp; C = C \\ &amp; / &amp; \diagdown \\ b &amp; &amp; b \end{array}, \begin{array}{c} a &amp; &amp; a \\ &amp; \diagdown &amp; / \\ &amp; C = N - OH \\ &amp; / &amp; \diagdown \\ b &amp; &amp; b \end{array}, \begin{array}{c} N &amp; &amp; N \\ &amp; \diagdown &amp; / \\ &amp; N = N \\ &amp; / &amp; \diagdown \\ a &amp; &amp; b \end{array}, \begin{array}{c} a &amp; &amp; a \\ &amp; \diagdown &amp; / \\ &amp; \text{Cyclohexane ring} \\ &amp; / &amp; \diagdown \\ b &amp; &amp; b \end{array}</math></p> <p> (Ring greater than 7 member with double bond), </p> <p>e.g. <math>\begin{array}{c} H &amp; &amp; H \\ &amp; \diagdown &amp; / \\ &amp; C = C \\ &amp; / &amp; \diagdown \\ HOOC &amp; &amp; COOH \end{array}</math> cis maleic acid</p> <p><math>\begin{array}{c} H &amp; &amp; COOH \\ &amp; \diagdown &amp; / \\ &amp; C = C \\ &amp; / &amp; \diagdown \\ HOOC &amp; &amp; H \end{array}</math> trans fumaric acid</p> |                   |
|         | (ii) Optical isomerism     | <p>d, l -isomerism</p> <p><math>\begin{array}{c} O \\    \\ H - C \\   \\ H - C - OH \\   \\ OH - C - OH \\   \\ H - C - OH \\   \\ H - C - OH \\   \\ CH_2OH \end{array}</math> D-Glucose</p> <p><math>\begin{array}{c} O \\    \\ H - C \\   \\ OH - C - H \\   \\ H - C - OH \\   \\ OH - C - H \\   \\ OH - C - H \\   \\ CH_2OH \end{array}</math> L-Glucose</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                   |

| Sr. No.       | Concept                                                                                                                                                                                                                                           | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Other information |            |                                   |                 |             |                                               |                                                         |              |           |                                               |                                                        |              |              |                                                         |                                                         |              |  |
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| 5.            | <b>Reaction mechanism &amp; intermediates:</b><br>(a) Fission of a covalent bond:<br>(i) Heterolytic cleavage<br><br>(ii) Homolytic cleavage<br><br>(b) Reaction intermediate                                                                     | $\text{H}_3\text{C} - \text{Br} \longrightarrow \text{}^+\text{CH}_3 + \text{Br}^-$<br>$\text{R} - \text{X} \longrightarrow \text{R}^\bullet + \text{X}^\bullet$ <p style="text-align: center;">Alkyl free radical</p> <table border="1"> <thead> <tr> <th>Intermediates</th><th>Definition</th><th>Hybridisation of carbon and shape</th><th>Stability order</th></tr> </thead> <tbody> <tr> <td>Carbocation</td><td>Carbon species carrying +ve charge on carbon.</td><td>sp<sup>2</sup> hybridised carbon and planar structure.</td><td>3° &gt; 2° &gt; 1°</td></tr> <tr> <td>Carbanion</td><td>Carbon species carrying -ve charge on carbon.</td><td>sp<sup>3</sup> hybridised carbon and pyramidal shape.</td><td>1° &gt; 2° &gt; 3°</td></tr> <tr> <td>Free radical</td><td>Atom or group of atoms having odd or unpaired electron.</td><td>sp<sup>2</sup> hybridised carbon and planar structure.</td><td>3° &gt; 2° &gt; 1°</td></tr> </tbody> </table> | Intermediates     | Definition | Hybridisation of carbon and shape | Stability order | Carbocation | Carbon species carrying +ve charge on carbon. | sp <sup>2</sup> hybridised carbon and planar structure. | 3° > 2° > 1° | Carbanion | Carbon species carrying -ve charge on carbon. | sp <sup>3</sup> hybridised carbon and pyramidal shape. | 1° > 2° > 3° | Free radical | Atom or group of atoms having odd or unpaired electron. | sp <sup>2</sup> hybridised carbon and planar structure. | 3° > 2° > 1° |  |
| Intermediates | Definition                                                                                                                                                                                                                                        | Hybridisation of carbon and shape                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Stability order   |            |                                   |                 |             |                                               |                                                         |              |           |                                               |                                                        |              |              |                                                         |                                                         |              |  |
| Carbocation   | Carbon species carrying +ve charge on carbon.                                                                                                                                                                                                     | sp <sup>2</sup> hybridised carbon and planar structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3° > 2° > 1°      |            |                                   |                 |             |                                               |                                                         |              |           |                                               |                                                        |              |              |                                                         |                                                         |              |  |
| Carbanion     | Carbon species carrying -ve charge on carbon.                                                                                                                                                                                                     | sp <sup>3</sup> hybridised carbon and pyramidal shape.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 1° > 2° > 3°      |            |                                   |                 |             |                                               |                                                         |              |           |                                               |                                                        |              |              |                                                         |                                                         |              |  |
| Free radical  | Atom or group of atoms having odd or unpaired electron.                                                                                                                                                                                           | sp <sup>2</sup> hybridised carbon and planar structure.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 3° > 2° > 1°      |            |                                   |                 |             |                                               |                                                         |              |           |                                               |                                                        |              |              |                                                         |                                                         |              |  |
| 6.            | <b>Attacking reagent:</b><br>(a) Nucleophile<br><br>(b) Electrophile                                                                                                                                                                              | <p>Nucleus seeking reagent which brings an electron pair.<br/>           e.g. <math>\text{}^\ominus\text{OH}</math>, <math>\text{}^\ominus\text{CN}</math></p> <p>Electron seeking reagent that takes away an electron pair.<br/> <math>&gt;\text{C}=\text{O}</math>, <math>\text{}^\oplus\text{Br}</math></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                   |            |                                   |                 |             |                                               |                                                         |              |           |                                               |                                                        |              |              |                                                         |                                                         |              |  |
| 7.            | <b>Electron displacement effects in covalent bonds:</b><br>(a) Inductive effect (I effect):<br>(i) -I effect:<br><br>(ii) +I effect:<br><br>(b) Electrometric effect<br>(i) +E effect<br>(ii) -E effect<br><br>(c) Resonance or Mesomeric effect: | <p>Displacement of <math>\sigma</math> electrons along a saturated carbon chain</p> <p>Electron withdrawing (-I effect)<br/> <math>-\text{NO}_2 &gt; -\text{CN} &gt; -\text{COOH} &gt; -\text{F} &gt; -\text{Cl} &gt; -\text{Br} &gt; -\text{I}</math></p> <p>Electron donating (+I effect)<br/> <math>(\text{CH}_3)_3\text{C} - &gt; (\text{CH}_3)_2\text{CH} - &gt; \text{CH}_3\text{CH}_2 - &gt; \text{C}_2\text{H}_5 - &gt; \text{CH}_3 -</math></p> <p>Transfer of <math>\pi</math> electrons of a multiple bond towards one of the bonded atoms.</p> <p>Electron transfer takes place towards the atom.</p> <p>Electron transfer takes place away from the atom</p> <p>Polarity produced in the molecule by the interaction of 2 <math>\pi</math> bonds or between a <math>\pi</math> bond &amp; lone pair of electrons.</p>                                                                                                                              |                   |            |                                   |                 |             |                                               |                                                         |              |           |                                               |                                                        |              |              |                                                         |                                                         |              |  |

| Sr. No. | Concept                                                                                                                                 | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Other information |
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|         | (i) +R or +M effect<br>(ii) -R or -M effect<br><br>(d) Hyperconjugation                                                                 | -OH, -OR, -OCOR, -SH, -SR, -NH <sub>2</sub> , -NHR, -NR <sub>2</sub> , -Cl, -Br, -I, etc.<br>-COOH, >C=O, -CHO, -COOR, -CN, -NO <sub>2</sub> , etc.<br><br>Delocalization of sigma electrons of C-H bond of an alkyl group directly attached to an atom of unsaturated system or to an atom with an unshared p-orbital.<br>CH <sub>3</sub> - > CH <sub>3</sub> CH <sub>2</sub> - > (CH <sub>3</sub> ) <sub>2</sub> CH - > (CH <sub>3</sub> ) <sub>3</sub> C -                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                   |
| 8.      | <b>Relative stability order:</b><br>(a) Stability of carbocation<br><br>(b) Stability of carbanion<br><br>(c) Stability of free radical | $(\text{CH}_3)_3\text{C}^+ > (\text{CH}_3)_2\text{CH}^+ > \text{CH}_3\text{CH}_2^+ > \text{CH}_3^+ > \text{HC}\equiv\text{C}^+ > \text{C}_6\text{H}_5^+$<br><br>$(\text{C}_6\text{H}_5)_3\text{C}^- > (\text{C}_6\text{H}_5)_2\text{CH}^- >> \text{C}_6\text{H}_5\text{CH}_2^- > \text{CH}_2=\text{CH}-\text{CH}_2^- > \text{HC}\equiv\text{C}^- > \text{H}_2\text{C}=\bar{\text{C}}\text{H} > \bar{\text{C}}\text{H}_3 > \text{CH}_3\bar{\text{C}}\text{H}_2 > (\text{CH}_3)_2\bar{\text{C}}\text{H} > (\text{CH}_3)_3\bar{\text{C}}$<br><br>$\text{CH}_2=\text{CH}\dot{\text{C}}\text{H}_2 > (\text{CH}_3)_3\dot{\text{C}} > (\text{CH}_3)_2\dot{\text{C}}\text{H} > \text{CH}_3\dot{\text{C}}\text{H}_2 > \dot{\text{C}}\text{H}_3$                                                                                                                                                                                                                                                                                                     |                   |
| 9.      | <b>Acidic strength</b>                                                                                                                  | <b>Acidic strength <math>\propto</math> Stability of conjugate base <math>\propto K_a \propto \frac{1}{pK_a}</math></b><br><br>(i) $\text{H}_2\text{O} > \text{CH}\equiv\text{CH} > \text{NH}_3$<br><br>(ii) $\text{CH}\equiv\text{CH} > \text{CH}_2=\text{CH}_2 > \text{CH}_3-\text{CH}_3$<br><br>(iii) $\text{R-SO}_3\text{H} > \text{R-COOH} > \text{C}_6\text{H}_5\text{OH} > \text{R-OH}$<br><br>(iv) $\text{HCOOH} > \text{CH}_3\text{COOH} > \text{CH}_3\text{CH}_2\text{COOH}$<br><br>(v) $\text{2,4,6-trinitrophenol} > \text{HCOOH} > \text{C}_6\text{H}_5\text{COOH} > \text{CH}_3\text{COOH}$<br><br>(vi) $\text{CCl}_3\text{COOH} > \text{CHCl}_2\text{COOH} > \text{CH}_2\text{ClCOOH}$<br><br>(vii) $\text{CH}_3-\text{CH}_2-\underset{\text{F}}{\text{CH}}-\text{COOH} > \text{CH}_3-\underset{\text{F}}{\text{CH}}-\text{CH}_2\text{COOH} > \text{CH}_2-\underset{\text{F}}{\text{CH}}-\text{CH}_2\text{COOH}$<br><br>(x) $\text{2,4,6-trinitrophenol} > \text{2,4-dinitrophenol} > \text{3-nitrophenol} > \text{phenol}$ |                   |

| Sr. No. | Concept        | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Other information |
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| 10.     | Basic strength | <p>Basic strength <math>\propto K_b \propto \frac{1}{pK_b}</math></p> <p>Basic strength of amine:</p> <p>In aqueous medium</p> <p><math>R \Rightarrow -CH_3 \quad 2^\circ &gt; 1^\circ &gt; 3^\circ &gt; NH_3</math></p> <p><math>R \Rightarrow -CH_2CH_3 \quad 2^\circ &gt; 3^\circ &gt; 1^\circ &gt; NH_3</math></p> <p>In gaseous medium</p> <p><math>R \Rightarrow -CH_3 \quad 3^\circ &gt; 2^\circ &gt; 1^\circ &gt; NH_3</math></p> <p><math>R \Rightarrow -CH_2CH_3 \quad 3^\circ &gt; 2^\circ &gt; 1^\circ &gt; NH_3</math></p> |                   |

